



Gut microbiota intervention by pre and probiotics can induce regulatory T cells and reduce the risk of severe acute GVHD following allogeneic hematopoietic stem cell transplantation

Ehsan Yazdandoust^a, Abbas Hajifathali^b, Elham Roshandel^b, Mahin Nikougoftar Zarif^c, Ali Akbar Pourfathollah^d, Sayeh Parkhideh^b, Mahshid Mehdizadeh^b, Sedigheh Amini-Kafiabad^{a,*}

^a Blood Transfusion Research Center, High Institute for Research and Education in Transfusion Medicine, Tehran, Iran

^b Hematopoietic Stem Cell Research Center, Shahid Beheshti University of Medical Sciences, Tehran, Iran

^c Center for Hematology and Regenerative Medicine, Karolinska Institutet, Department of Medicine, Karolinska University Hospital Huddinge, Stockholm, Sweden

^d Department of Immunology, Faculty of Medical Sciences, Tarbiat Modares University, Tehran, Iran

ARTICLE INFO

Keywords:

Hematopoietic stem cell transplantation
Acute graft-versus-host disease (aGVHD)
Regulatory T cell
Prebiotic
Probiotic

ABSTRACT

Background: Acute graft-versus-host disease (aGVHD) is one of the leading causes of limitation and mortality after allogeneic hematopoietic stem cell transplantation (allo-HSCT). Numerous studies have shown that changes in the gut microbiome diversity increased post-transplant problems, including the occurrence of aGVHD. Probiotics and prebiotics can reconstitute the gut microbiota and thus increase bacterial metabolites such as short-chain fatty acids (SCFAs) that have immunomodulatory effects preventing aGVHD in recipients of allo-HSCTs.

Methods/Study Design: We conducted a pilot randomized clinical trial to investigate whether oral synbiotics are associated with the prevention or reduction in occurrence/severity and mitigate complications of aGVHD following allo-HSCT. A commercially available synbiotic mixture containing high levels of 7 safe bacterial strains plus fructo-oligosaccharides as a prebiotic was administered to allo-HSCT recipients. Out of 40 allo-HSCT patients, 20 received daily a synbiotic 21 days prior to transplantation (days -21 to day 0). In contrast, in the control group 20 recipients of allo-HSCT did not receive a symbiotic therapy.

Results: Within first 100 days of observation, the incidence of severe (grade III/IV) aGVHD in the a synbiotic-therapy group was 0% (0 out of 20 patients), whereas it was 25% (5 out of 20 patients) in the control group ($P = 0.047$). The median percentage of CD4 + CD25 + Foxp3+ regulatory T cells (Tregs) among CD4+ lymphocytes on day 28 after HSCT in the synbiotic group was higher (2.54%) than in control group (1.73%; $P = 0.01$). There was no difference in Treg cells on day 7 after HSCT between two groups. However, the median percentage and the absolute count of Tregs in patients who experience aGVHD was significantly lower on days 7 and 28 after HSCT (both $P < 0.05$). The overall 12-month survival (OS) rate was higher (90%) in the symbiotic-treated patients than in the control group (75%), but the difference was not statistically significant ($P = 0.234$).

Conclusion: Our preliminary findings suggest that synbiotic intake before and during the conditioning regimen of allo-HSCT patients may lead to a reduction in the incidence and severity of aGVHD through the induction of CD4 + CD25 + Foxp3+ regulatory T cells, thus contributing to the improvement of transplant outcomes. Much larger studies are needed to confirm our observations.

1. Introduction

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is a leading immunotherapeutic approach and a potential cure for hematological malignancies. Despite its increasing adoption worldwide, graft-

versus-host disease (GVHD) is a significant concern accompanying the allo-HSCT procedure. It is linked to higher mortality rates and damage to the gastrointestinal tract and liver or skin [1,2].

In light of the increasing knowledge of the gut microbiota ecosystem, various studies and publications have shown that alterations and

* Corresponding author.

E-mail address: dr.amini@gmail.com (S. Amini-Kafiabad).

<https://doi.org/10.1016/j.trim.2023.101836>

Received 2 January 2023; Received in revised form 2 April 2023; Accepted 5 April 2023

Available online 8 April 2023

0966-3274/© 2023 Elsevier B.V. All rights reserved.

decreases in gut microbiome diversity can lead to post-transplant complications, such as acute graft-versus-host disease (aGVHD), acute gastrointestinal problems, and intestinal infections [3–5]. It was also shown that the diversity of normal intestinal flora remains a crucial determinant of the favorable outcomes in allo-HSCT. Patients exhibiting reduced diversity in intestinal flora had poorer overall survival rates than those with higher diversity [6].

Probiotics and prebiotics reconstitute normal gastrointestinal bacterial flora, maintain normal intestinal bacterial flora, and potentially prevent the growth of opportunistic bacteria that predispose patients to GVHD. Probiotics may reduce the occurrence/severity of GVHD by inducing regulatory T cells (Tregs), which release anti-inflammatory cytokines and regulate the host immune system [4,7,8].

The normal intestinal microbial flora produces short-chain fatty acids (SCFAs), predominantly butyrate, which suppresses the NF- κ B signaling pathway, increases the expression of IL-10, and also affects macrophages and dendritic cells [9,10]. These compounds modulate immune responses in the gut and may reduce the occurrence or severity of GVHD by inducing Tregs [11–13]. Also, it has been demonstrated that butyrate-producing bacteria mitigate aGVHD severity in murine models [11].

The present clinical trial addressed the effects of oral synbiotics, composed of both probiotic and prebiotic components, on GVHD after allo-HSCT. The study aimed to assess the impact of probiotics and prebiotics on various parameters such as Treg induction, GVHD prevention, overall survival rates, and intestinal complications in the participants.

2. Methods/study design

2.1. Study design and objectives

The current clinical trial was conducted on patients undergoing myeloablative allo-HSCT between October 2019 and February 2022 at the Taleghani Educational Hospital, Tehran, Iran. The present study examined whether synbiotic supplementation could reduce acute GVHD occurrence/severity in allo-HSCT patients. Forty patients were enrolled in the study, and informed consent was obtained from all participants. The clinical trial was registered at fa.ict.ir with registration number IRCT20190914044766N1 and was approved by the Ethics Committees of the High Institute for Education and Research in Transfusion Medicine (ethic code: IR.TMI.REC.1398.023) and the Ethics Committees of Shahid Beheshti University of Medical Sciences (ethic code: IR.SBMU.REC.1399.003). The intervention group comprised 20 patients who received a synbiotic supplementation capsule (Famifact, Zist Takhmir, Tehran, Iran) daily after lunch from day –21 to day 0 (the transplant day). The control group comprised 20 patients who did not receive any probiotics. Each synbiotic capsule contained a high level (10^9 CFU) of seven bacterial strains that are safe and beneficial to the human body, plus fructooligosaccharides (FOS) as a prebiotic. The bacterial strains used were *Lactobacillus rhamnosus*, *Lactobacillus casei*, *Lactobacillus bulgaricus*, *Lactobacillus acidophilus*, *Bifidobacterium breve*, *Bifidobacterium longum*, and *Streptococcus thermophilus*. Patient assignment to the study groups was accomplished by utilizing the block randomization method. Specifically, an online software tool available at <https://www.sealedenvelope.com/simple-randomiser/v1/lists> was implemented to generate a random number utilizing a block size of 4, 6 or 8. All patients were monitored for up to 12 months after transplantation, and their survival and occurrence or grade of acute GVHD were assessed according to the established criteria [14].

2.2. Patient selection

All participants included in this study were between 20 and 50 years of age and with cytokine-mobilized peripheral blood stem cells (allo-PBSC) as the source of stem cells. Eligible donors were either 10/10 full HLA-matched as determined by high-resolution typing at the HLA-A,

–B, –C, –DRB1, and –DQB1 loci or had a single disparity (9/10 match). Prophylactic treatment of graft-versus-host-disease (GVHD) consisted of cyclosporine A (CsA A; Sandoz), administered at a dosage of 3 mg/kg/day IV from days –2 to +5 of transplantation and IV methotrexate (MTX; Sandoz) administered at a dose of 10 mg/m² on day +1 and 6 mg/m² on days +3, +6, and day +11. All patients received irradiated blood components. The exclusion criteria for the patients during the intervention included (1) blood infection and severe diarrhea, (2) occurrence of fever above 38.5 degrees, (3) allergy to any components of the synbiotic product, and (4) self-prescribed probiotics and prebiotic therapy.

2.3. Sample size

We utilized G*power software to determine the appropriate sample size for our study. This involved adopting a large effect size ($f = 0.4$), setting the significance level at 0.05 and a power of 80%. As a result of these calculations, a total of 40 patients were necessary to achieve adequate statistical power.

2.4. Regulatory T cell evaluation

For the assessment of CD4 + CD25 + Foxp3+ Tregs, fresh EDTA blood samples were collected from patients at two-time points, day +7 and day +28 post-transplantation. Multi-color flow cytometry was employed to analyze peripheral blood mononuclear cells (PBMCs) isolated by Ficoll-Hypaque density gradient. The FlowXTM Human Regulatory T Cell Multi-Color Flow Cytometry Kit (R&D Systems, USA) was used for staining with three fluorochrome-conjugated antibodies, following the manufacturer's instructions. The lymphocytes were first stained with Human CD4-FITC (Mouse IgG2A; Clone 11830) and Human IL-2 R/CD25-APC (Mouse IgG2A; Clone 24212) and then permeabilized with the kit's Permeabilization Buffer. After permeabilization, 10 μ L of Human/Mouse FoxP3-PE (Rabbit IgG; Clone 1054C) or the rabbit IgG-PE isotype control included in this kit was added to the cells and then incubated for 30 min at 2–8 °C. The cells were washed, and the readout was performed using the AttuneTM NxT Flow Cytometer (Thermo Fisher Scientific, USA). The gating strategy for the Treg cell population was First, lymphocytes were identified according to FSC height to SSC height signal distribution (Lymphocyte gate), then CD4+ lymphocytes were gated, and next the median percentage of CD4+ FoxP3+ CD25+ cells was evaluated in the peripheral blood (Fig. 1).

2.5. Clinical monitoring of patients

Throughout the duration of this study, we conducted a comprehensive follow-up on each of the participating patients for a maximum period of 12 months. During this time, we scrutinized and evaluated the emergence of positive blood cultures and infections, severe diarrhea, incidence of mucositis, overall survival statistics, and the occurrence or grading of acute GVHD by adhering to the norms outlined in the consensus conference on acute GVHD grading [14]. In addition, we employed the Common Terminology Criteria for Adverse Events (version 3) to document and report any and all instances of adverse events (AEs).

2.6. Statistical analysis

Quantitative variables were presented as either mean and standard deviation or median and interquartile range based on the results of the normality test conducted by the Shapiro test. Descriptive statistics such as frequency and percentage were applied to report quantitative variables. The independent samples *t*-test was utilized to evaluate the difference between the means of two independent groups, while the paired sample *t*-test was used to assess the change of variables over time. Furthermore, the chi-square test with continuity correction was used to

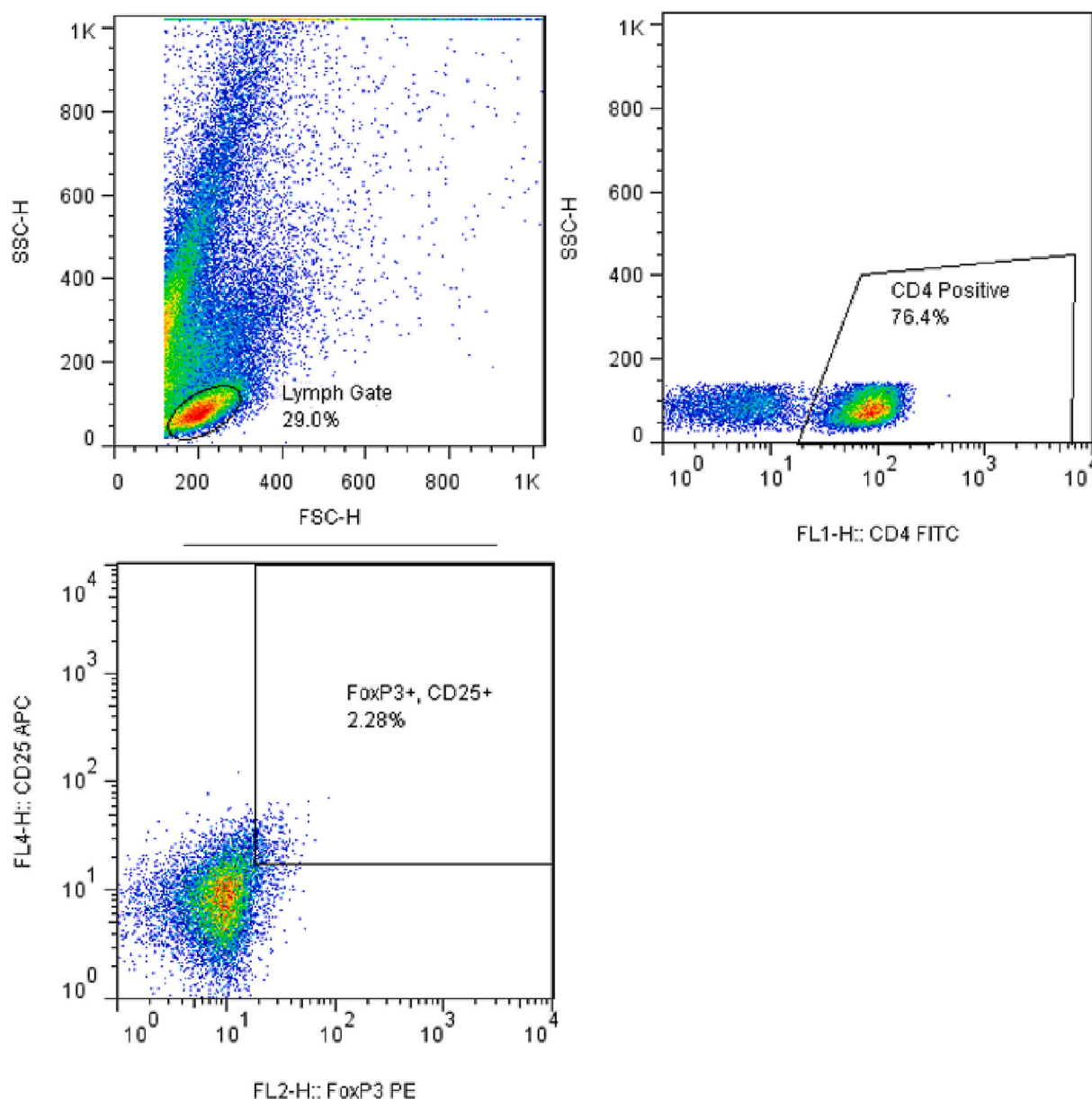


Fig. 1. A representative illustration of regulatory T cells gating.

determine the association between two qualitative variables, and the Fisher exact test was used if needed. Additionally, the Cox regression model was implemented to estimate the hazard ratio, while the Kaplan-Meier method was used to estimate survival probability. The study's statistical significance level was set at 0.05, and all statistical analysis was performed using R software (version 4).

3. Results

3.1. Patient characteristics

A total of forty patients were enrolled in the trial and divided into two groups, with twenty patients receiving daily synbiotic capsules and the remaining twenty receiving no probiotics or prebiotics. The demographic and clinical characteristics of all participants were subsequently summarized in Table 1. The proportion of males to females and the degree of HLA compatibility were almost similar between the two groups, with no significant differences observed in GVHD prophylaxis regimens (P -value = 1), conditioning regimen (P -value = 0.79),

prophylactic antibiotic drugs (P -value = 0.66), and other pertinent parameters outlined in Table 1.

3.2. Safety analysis

Among twenty patients assigned to the intervention group, none exhibited any episodes of bacteremia with *Lactobacillus rhamnosus*, *Lactobacillus casei*, *Lactobacillus bulgaricus*, *Lactobacillus acidophilus*, *Bifidobacterium breve*, *Bifidobacterium longum*, and *Streptococcus thermophilus*. Further, the intervention group did not experience any adverse events, such as infection, bacteremia, or any other unexpected complications related to our intervention compared to the control group, as depicted in Table 2.

3.3. Evaluation of acute GVHD

Until day +100, two patients in the intervention group experienced aGVHD (accounting for only 10% of the group population), while eight patients in the control group experienced acute GVHD (40% prevalence)

Table 1
Patient characteristics.

		Probiotics (N = 20)	Control (N = 20)	P- value
Sex	Male	13(65%)	12(60%)	0.744
	Female	7(35%)	8(40%)	
Age (years)	Mean $\pm$ SD	38.5 $\pm$ 10.3	34.9 $\pm$ 9.8	0.269
	Range	20–50	20–50	
Diagnosis	AML	7 (35%)	10(50%)	0.355
	ALL	9(45%)	7(35%)	
	NHL	1(5%)	3(15%)	
Stem cell source	MDS	2(10%)	0 (0.00)	1.00
	Myelofibrosis	1(5%)	0 (0.00)	
	PBSCs	20(100%)	20(100%)	
Donor sex	Male	15	15	1.00
	Female	5	5	
Donating Female-male	Yes	2(10%)	3(15%)	1.00
	No	18(90%)	17(85%)	
Donor type, n (%)	Matched related	16(80%)	19(95%)	0.342
	Matched unrelated	3(15%)	0	
	Mismatched unrelated	1(5%)	1(5%)	
	Quinolone (levofloxacin/Ciprofloxacin)	15	12	
Prophylactic antibiotic drugs	carbapenem	3	4	0.66
	Intravenous vancomycin	5	8	
	Cyclosporine base	20(100%)	20(100%)	
GVHD prophylaxis	CSA + MTX	20(100%)	20(100%)	1.00
	Bu/Cy	12	11	
Conditioning regimens	Bu/Flu	8	9	0.79

AML, acute myeloid leukemia; ALL, acute lymphoblastic leukemia; Bu, busulfan; Cy, cyclophosphamide; CSA, cyclosporine A; Flu, fludarabine; PBSCs, peripheral blood stem cell; NHL, non-Hodgkin's lymphoma; MDS, myelodysplastic syndromes; MTX, methotrexate.

Table 2
Safety and Clinical outcomes.

	Probiotics (N = 20)	Control (N = 20)	P
Serious adverse event by day 28	1	1	1.00
High-grade mucositis	1	2	1.00
Bacteremia caused by bacteria in the probiotic capsule	0	0	1.00
Bacteremia by day + 28 with other bacteria	2 (10%)	3 (15%)	1.00
Diarrhea grade 2 and 3	2 (10%)	2 (10%)	1.00
Severe diarrhea grade 4	0	0	1.00
Total occurrence of acute GVHD	2 (10%)	8 (40%)	0.067
severe (grade III-IV) acute GVHD	0 (0%)	5 (25%)	0.047
Grade 0 acute GVHD	18 (90%)	12 (60%)	0.067
Grade 1 acute GVHD	1 (5%)	3 (15%)	
Grade 2 acute GVHD	1 (5%)	0 (0%)	
Grade 3 acute GVHD	0 (0%)	4 (20%)	
Grade 4 acute GVHD	0 (0%)	1 (5%)	
One-year overall survival	18 (90%)	15 (75%)	0.234
1-year chronic GVHD	4 (20%)	9 (45%)	0.177
Severe chronic GVHD	2 (10%)	4 (20%)	0.661
Relapse of the underlying disease	1 (5%)	2 (10%)	1.00

(P -value = 0.068, RR = 4; 95%CI: 0.97–16.5). In the intervention group, the cumulative incidence of severe (grade III or IV) acute GVHD was 0%, whereas it was 25% in the control group (P = 0.047) (Table 2). The median duration of aGVHD manifestation was approximately 41 days (range: 18–100) following the transplantation day. Among patients diagnosed with acute GVHD, five developed skin and gastrointestinal (GI) GVHD; two cases had skin GVHD; one case developed hepatic-GI GVHD, and one had hepatic GVHD.

3.4. Regulatory T cells in the peripheral blood of patients

The median percentage and absolute count of Tregs among CD4+ lymphocytes were determined on day +7 and day +28 after HSCT. The results indicate that there was no statistically significant difference between the control and intervention groups concerning Treg levels on day +7 after HSCT $1.62 \pm 1.17\%$ ($1.48 \pm 1.6/\mu\text{l}$) and $1.41 \pm 0.84\%$ ($1.38 \pm 1.2/\mu\text{l}$), respectively; P -value = 0.514). However, on day +28 after HSCT, there was a significantly higher percentage and absolute count of Tregs in the intervention group as compared to the control group $2.54 \pm 0.94\%$ ($3.12 \pm 1.41/\mu\text{l}$) and $1.73 \pm 0.95\%$ ($1.9 \pm 1.2/\mu\text{l}$), respectively; P -value = 0.01), as reported in Table 3 and illustrated in Fig. 2. Furthermore, it was observed that the median percentage and absolute count of Tregs in patients who experienced acute GVHD was significantly lower on days +7 ($0.7/\mu\text{l}$ vs. $1.62/\mu\text{l}$, P -value = 0.027) and + 28 ($1.65/\mu\text{l}$ vs. $3.9/\mu\text{l}$, P -value = 0.007) after HSCT (Fig. 3).

3.5. Overall survival

In this study, the overall survival rates were comparatively higher in patients who received probiotic interventions. The 1-year overall survival rates in the intervention and the control groups were reported as 90% and 75%, respectively, yet the difference was not statistically significant (P -value = 0.234) (Fig. 4). Furthermore, the hazard ratio for death was 2.584 (CI: 0.5–13.3, P -value = 0.257) in the control group. Additionally, the intervention group experienced a lower incidence of 1-year non-relapse mortality (5%) than the control group (15%). Out of the patients in the control group, three died due to acute and chronic GVHD, while two died due to the relapse of their underlying disease. In contrast, one patient in the intervention group died of relapse, and another died due to secondary graft failure.

4. Discussion

Our findings in this clinical trial study have revealed that using synbiotics daily, before and during the conditioning regimen for patients undergoing allo-HSCT, could lower the frequency and intensity of aGVHD and increase the induction of CD4+ CD25+ FOXP3+ Tregs post-transplantation. Corresponding studies have also demonstrated that the intestinal microbiome can produce various metabolites, such as SCFAs, which can affect the host immune system and facilitate the induction of Tregs [12,15,16].

Several studies have shown that higher levels of Tregs in allo-HSCT patients abate the risk of GVHD, whereas diminished levels of Tregs amplify the risk of GVHD [17–21]. In certain other studies, expanding Tregs have been employed to forestall and treat GVHD [22–25]. Interestingly, our study demonstrated that patients who experienced acute GVHD had lower Tregs. Furthermore, the group of patients who received synbiotics had a higher level of Tregs relative to the control cohort, thus signifying that synbiotics potentially assume a significant role in the in vivo expansion and induction of Tregs and the prevention of aGVHD.

Using probiotic and prebiotic therapy to reconstruct microbiota has emerged as a promising approach for mitigating the severity of GVHD and its related hazards [8,26,27]. Moreover, this strategy has demonstrated significant potential in providing resistance against intestinal pathogens, particularly *Clostridium difficile* infection [28].

The present pilot study in patients undergoing hematopoietic stem

Table 3
Regulatory T Cell in the peripheral blood of patients on day +7 and day +28 after HSCT.

	Time	Control	Intervention	P-value
Treg %	Day +7	1.62 ± 1.17	1.41 ± 0.84	0.514
	Day +28	1.73 ± 0.95	2.54 ± 0.94	0.01
	P-value	0.751	<0.001	

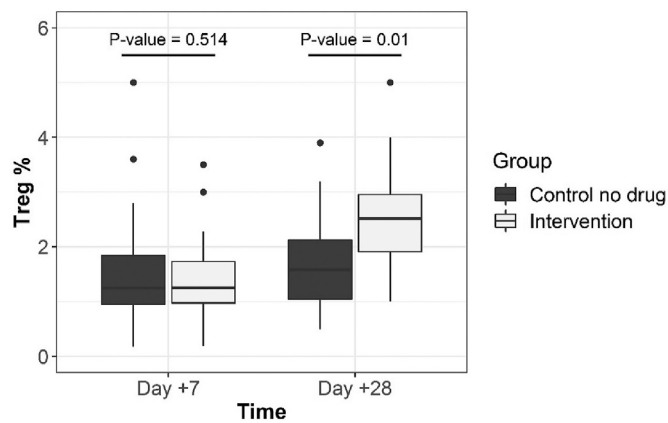


Fig. 2. Percentage of Treg level in the peripheral blood among control group versus probiotic group (intervention group) on day +7 and day +28 after HSCT.

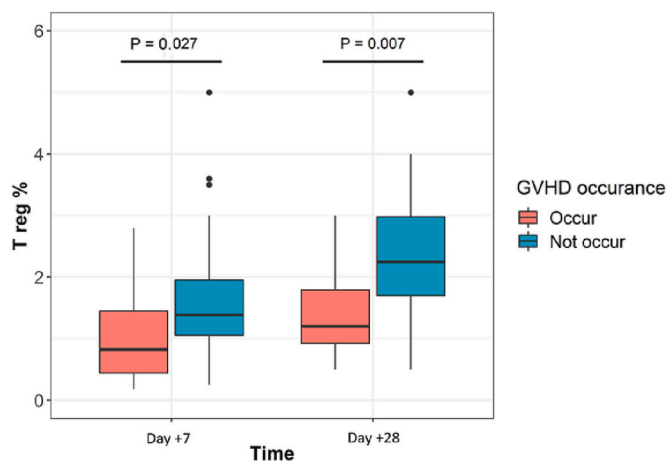


Fig. 3. Percentage of Treg levels in peripheral blood on days +7 and +28 after HSCT in patients who experienced aGVHD in comparison to those who did not experience aGVHD.

cell transplantation (HSCT) represents a pioneering effort to combine FOS as a prebiotic substrate and probiotics with multiple bacterial strains. The results of the trial suggest that this combination is safe in the pre-transplant phase as well as throughout the conditioning period. Before this, only one strain of the probiotic *Lactobacillus plantarum* (LBP) and one FOS prebiotic, respectively, were utilized in the studies by Ladas [29] and Andermann [30].

A research study conducted by Biagi et al. (2019) demonstrated that allo-HSCT patients with low intestinal bacterial diversity even before transplantation, low levels of *Blautia* bacteria, and the presence of *Fusobacterium* opportunistic pathogen are at increased risk of GVHD [31]. Similarly, using probiotics and prebiotics in our study before the transplant process and during the conditioning period ameliorated potential complications associated with allo-HSCT, including GVHD and infection with opportunistic pathogens, highlighting the potential benefit of synbiotics in elevating intestinal bacterial diversity.

In a study by Gorshein et al. in 2017, the efficacy of *Lactobacillus rhamnosus* probiotic in inhibiting GVHD was demonstrated in mice. However, contrary to the murine findings, a clinical trial by the same authors consisting of 31 patients, wherein 20 patients were given a *Lactobacillus rhamnosus* probiotic while 11 received no treatment, showed insignificant outcomes in the prevention of GVHD [27,32]. The lack of efficacy in their clinical trial may have been due to the small sample size, uneven distribution of participants across the treatment and control groups, and using probiotics with only one bacterial strain. However, our investigation revealed that implementing a probiotic regimen comprising a collection of seven advantageous bacterial strains and fructooligosaccharides (FOS), which amplifies the diversity of intestinal bacteria exhibited effective performance in mitigating graft-versus-host disease (GVHD). This outcome emphasizes the prospective application of this approach in preventing GVHD.

Several studies have been conducted to augment gut microbiota diversity in the context of Hematopoietic Stem Cell Transplantation (HSCT). These endeavors include the work of Taur et al., who employed autologous fecal microbiota transplantation [3], Andermann et al., who utilized the FOS prebiotic for the same purpose [30], and Yoshifuji et al., who evaluated the efficacy of commercially available prebiotics mixture consisting of glutamine, fiber, and oligosaccharide (GFO) [8].

The research of Taur et al. demonstrated the substantial significance of diversity within the intestinal microbiome [33]. The study found that

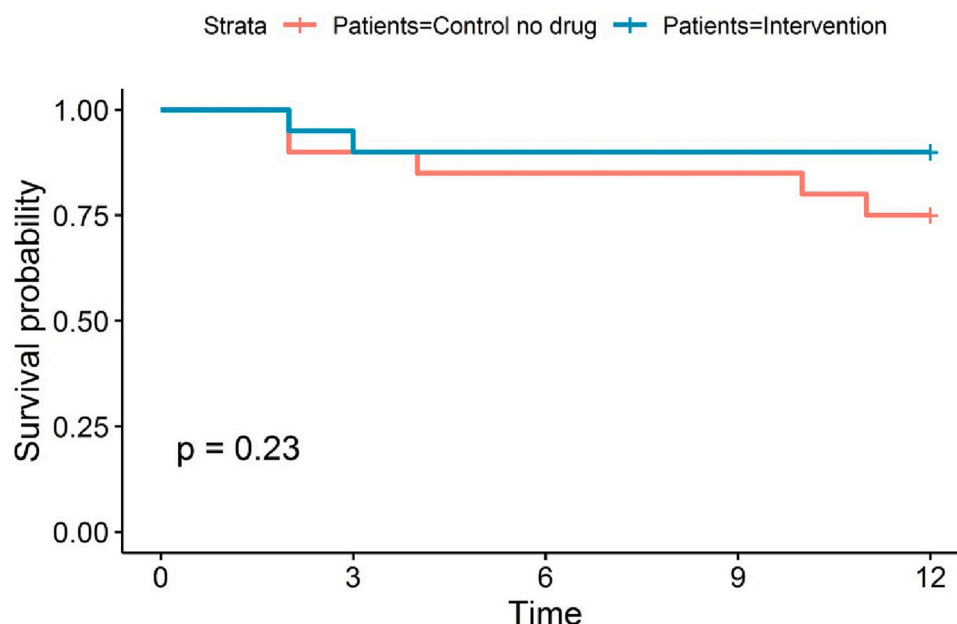


Fig. 4. Survival curve in terms of patient groups. The curve and P-value are estimated by the Kaplan Meier method.

patients with low bacterial diversity had significantly lower overall survival rates than those with moderate to high diversity. Specifically, patients with low and moderate diversity had 3-year overall survival rates of 36% and 60%, respectively, while those with high bacterial diversity had a 3-year overall survival rate of 67%. This finding was similarly observed in the study conducted by Staf et al. [34]. The present study also observed comparable results where patients who consumed probiotics exhibited an elevated one-year survival rate. This outcome could be attributed to the effect of probiotics on expanding the diversity of the intestinal microbiome and minimizing the likelihood of severe GVHD occurrence.

The present pilot study was subject to certain limitations, such as the absence of an assessment of gut microbiome diversity. However, according to a parallel, double-blind, controlled, randomized clinical trial, individuals who received the Familiact synbiotic, the same intervention used in our study, exhibited an increased diversity of gut microbes following the intervention. This increase was quantified using the sequencing method [35]. Furthermore, several prior studies have also reported a significant increase in gut microbiota composition and diversity following the synbiotic intervention [36–39]. Nonetheless, the primary objective of our research was to examine the preventive effects of synbiotics on GVHD incidence and their impact on both Treg cell numbers and overall survival rates within the context of a randomized pilot study. Additional well-designed investigations into the consequences of synbiotic use on hematopoietic stem cell transplantation outcomes are deemed necessary.

Finally, the administration of probiotics plus prebiotics as an intervention before and during the conditioning regimens in patients undergoing allo-HSCT can potentially lower the incidence and severity of GVHD through the induction of regulatory T cells, thus contributing to the improvement of transplant outcomes.

Funding

This work was supported by the High Institute for Research and Education in Transfusion Medicine. This was also supported by Iran National Science Foundation (INSF).

Declaration of Competing Interest

The authors declare no conflict of interest.

Data availability

Data will be made available on request.

Acknowledgments

We thank all the patients who participated in this study and the nurses and physicians of Taleghani Educational Hospital, Tehran, Iran.

We would like to express appreciation and thanks to Dr. Amir Teimourpour for statistical assistance.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.trim.2023.101836>.

References

- [1] P.J. Martin, G. Schoch, L. Fisher, V. Byers, C. Anasetti, F.R. Appelbaum, et al., A retrospective analysis of therapy for acute graft-versus-host disease: initial treatment, *Blood*. 76 (8) (1990) 1464–1472.
- [2] K. Sullivan, M.T. Mori, J. Sanders, M. Siadak, R. Witherspoon, C. Anasetti, et al., Late complications of allogeneic and autologous marrow transplantation, *Bone Marrow Transplant*. 10 (Suppl. 1) (1992) 127–134.
- [3] Y. Taur, K. Coyte, J. Schluter, E. Robilotti, C. Figueroa, M. Gjonbalaj, et al., Reconstitution of the gut microbiota of antibiotic-treated patients by autologous fecal microbiota transplant, *Sci. Transl. Med.* 10 (460) (2018) eaap9489.
- [4] F. Noor, A. Kaysen, P. Wilmes, J.G. Schneider, The gut microbiota and hematopoietic stem cell transplantation: challenges and potentials, *J. Innate Immun.* 11 (5) (2019) 405–415.
- [5] S. Murphy, V.H. Nguyen, Role of gut microbiota in graft-versus-host disease, *Leuk. Lymphoma* 52 (10) (2011) 1844–1856.
- [6] Pamer E.G.; Taur Y.; Jenq R.; van den Brink M.R.; et al., *Blood*. 124. SCI–48.
- [7] K. Yoshioka, K. Kakihana, N. Doki, K. Ohashi, Gut microbiota and acute graft-versus-host disease, *Pharmacol. Res.* 122 (2017) 90–95.
- [8] K. Yoshifuji, K. Inamoto, Y. Kiridoshi, K. Takeshita, S. Sasajima, Y. Shiraishi, et al., Prebiotics protect against acute graft-versus-host disease and preserve the gut microbiota in stem cell transplantation, *Blood Adv.* 4 (19) (2020) 4607–4617.
- [9] P.V. Chang, L. Hao, S. Offermanns, R. Medzhitov, The microbial metabolite butyrate regulates intestinal macrophage function via histone deacetylase inhibition, *Proc. Natl. Acad. Sci. U. S. A.* 111 (6) (2014) 2247–2252.
- [10] M.A. Vinolo, H.G. Rodrigues, R.T. Nachbar, R. Curi, Regulation of inflammation by short chain fatty acids, *Nutrients*. 3 (10) (2011) 858–876.
- [11] N.D. Mathewson, R. Jenq, A.V. Mathew, M. Koenigsnecht, A. Hanash, T. Toubai, et al., Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease, *Nat. Immunol.* 17 (5) (2016) 505–513.
- [12] K. Atarashi, T. Tanoue, K. Oshima, W. Suda, Y. Nagano, H. Nishikawa, et al., Treg induction by a rationally selected mixture of Clostridia strains from the human microbiota, *Nature*. 500 (7461) (2013) 232–236.
- [13] Y. Furusawa, Y. Obata, S. Fukuda, T.A. Endo, G. Nakato, D. Takahashi, et al., Commensal microbe-derived butyrate induces the differentiation of colonic regulatory T cells, *Nature*. 504 (7480) (2013) 446–450.
- [14] D. Przemyk, D. Weisdorf, P. Martin, H.G. Klingemann, P. Beatty, J. Hows, et al., 1994 consensus conference on acute GVHD grading, *Bone Marrow Transplant*. 15 (6) (1995) 825–828.
- [15] P.M. Smith, M.R. Howitt, N. Panikov, M. Michaud, C.A. Gallini, Y.M. Bohlooly, et al., The microbial metabolites, short-chain fatty acids, regulate colonic Treg cell homeostasis, *Science*. 341 (6145) (2013) 569–573.
- [16] M.D. Docampo, J.J. Auletta, R.R. Jenq, Emerging influence of the intestinal microbiota during allogeneic hematopoietic cell transplantation: control the gut and the body will follow, *Biol Blood Marrow Transplant*. 21 (8) (2015) 1360–1366.
- [17] J.S. Whangbo, J.H. Antin, J. Koreth, The role of regulatory T cells in graft-versus-host disease management, *Expert. Rev. Hematol.* 13 (2) (2020) 141–154.
- [18] K. Rezvani, S. Mielke, M. Ahmadzadeh, Y. Kilical, B.N. Savani, J. Zeilah, et al., High donor FOXP3-positive regulatory T-cell (Treg) content is associated with a low risk of GVHD following HLA-matched allogeneic SCT, *Blood*. 108 (4) (2006) 1291–1297.
- [19] J.M. Maganau, X. Qin, I. Tawara, C.E. Rogers, C. Kitko, M. Schlough, et al., Frequency of CD4(+)CD25(hi)FOXP3(+) regulatory T cells has diagnostic and prognostic value as a biomarker for acute graft-versus-host-disease, *Biol Blood Marrow Transplant*. 16 (7) (2010) 907–914.
- [20] R.D. Danby, W. Zhang, P. Medd, T.J. Littlewood, A. Peniket, V. Rocha, et al., High proportions of regulatory T cells in PBSC grafts predict improved survival after allogeneic haematopoietic SCT, *Bone Marrow Transplant*. 51 (1) (2016) 110–118.
- [21] S.N. Ukena, J. Grosse, E. Mischak-Weissinger, S. Buchholz, M. Stadler, A. Ganser, et al., Acute but not chronic graft-versus-host disease is associated with a reduction of circulating CD4(+)CD25 (high)CD127 (low/–) regulatory T cells, *Ann. Hematol.* 90 (2) (2011) 213–218.
- [22] B. Del Papa, L. Ruggeri, E. Urbani, S. Baldoni, D. Cecchini, T. Zei, et al., Clinical-grade-expanded regulatory T cells prevent graft-versus-host disease while allowing a powerful T cell-dependent graft-versus-leukemia effect in murine models, *Biol Blood Marrow Transplant*. 23 (11) (2017) 1847–1851.
- [23] C. Riegel, T.J. Boeld, K. Doser, E. Huber, P. Hoffmann, M. Edinger, Efficient treatment of murine acute GvHD by in vitro expanded donor regulatory T cells, *Leukemia*. 34 (3) (2020) 895–908.
- [24] J.M. Coghill, M.J. Carlson, T.P. Moran, J.S. Serody, The biology and therapeutic potential of natural regulatory T-cells in the bone marrow transplant setting, *Leuk. Lymphoma* 49 (10) (2008) 1860–1869.
- [25] J. Lu, P. Li, X. Du, Y. Liu, B. Zhang, F. Qi, Regulatory T cells induce transplant immune tolerance, *Transpl. Immunol.* 67 (2021), 101411.
- [26] R.R. Jenq, C. Ubeda, Y. Taur, C.C. Menezes, R. Khanin, J.A. Dudakov, et al., Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation, *J. Exp. Med.* 209 (5) (2012) 903–911.
- [27] E. Gorshein, C. Wei, S. Ambrosy, S. Budney, J. Vivas, A. Shenkerman, et al., Lactobacillus rhamnosus GG probiotic enteric regimen does not appreciably alter the gut microbiome or provide protection against GVHD after allogeneic hematopoietic stem cell transplantation, *Clin. Transpl.* 31 (5) (2017).
- [28] C.G. Buffie, I. Jarchum, M. Equinda, L. Lipuma, A. Gobourne, A. Viale, et al., Profound alterations of intestinal microbiota following a single dose of clindamycin results in sustained susceptibility to Clostridium difficile-induced colitis, *Infect. Immun.* 80 (1) (2012) 62–73.
- [29] E. Ladas, M. Bhatia, L. Chen, E. Sandler, A. Petrovic, D. Berman, et al., The safety and feasibility of probiotics in children and adolescents undergoing hematopoietic cell transplantation, *Bone Marrow Transplant*. 51 (2) (2016) 262.
- [30] T.M. Andermann, F. Fouladi, F.B. Tamburini, B. Sahaf, E. Tkachenko, C. Greene, et al., A fructo-oligosaccharide prebiotic is well tolerated in adults undergoing allogeneic hematopoietic stem cell transplantation: a phase I dose-escalation trial, *Transplant Cell Ther.* 27 (11) (2021) (932.e1–e11).

- [31] E. Biagi, D. Zama, S. Rampelli, S. Turrone, P. Brigidi, C. Consolandi, et al., Early gut microbiota signature of aGvHD in children given allogeneic hematopoietic cell transplantation for hematological disorders, *BMC Med. Genet.* 12 (1) (2019) 49.
- [32] A. Gerbitz, M. Schultz, A. Wilke, H.-J. Linde, J. Schölmerich, R. Andreesen, et al., Probiotic effects on experimental graft-versus-host disease: let them eat yogurt, *Blood.* 103 (11) (2004) 4365–4367.
- [33] Y. Taur, R.R. Jenq, M.A. Perales, E.R. Littmann, S. Morjaria, L. Ling, et al., The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation, *Blood.* 124 (7) (2014) 1174–1182.
- [34] A. Staffas, M. Burgos da Silva, M.R. van den Brink, The intestinal microbiota in allogeneic hematopoietic cell transplant and graft-versus-host disease, *Blood.* 129 (8) (2017) 927–933.
- [35] F. Zeinali, S.M. Aghaei Zarch, M.Y. Vahidi Mehrjardi, S.M. Kalantar, A. Jahan-Mihan, E. Karimi-Nazari, et al., Effects of synbiotic supplementation on gut microbiome, serum level of TNF- α , and expression of microRNA-126 and microRNA-146a in patients with type 2 diabetes mellitus: study protocol for a double-blind controlled randomized clinical trial, *Trials.* 21 (1) (2020) 324.
- [36] I.N. Sergeev, T. Aljutaily, G. Walton, E. Huarte, Effects of synbiotic supplement on human gut microbiota, body composition and weight loss in obesity, *Nutrients.* 12 (1) (2020) 222.
- [37] S.H. Lee, H.S. You, H.-G. Kang, S.S. Kang, S.H. Hyun, Association between altered blood parameters and gut microbiota after synbiotic intake in healthy, elderly Korean women, *Nutrients.* 12 (10) (2020) 3112.
- [38] N. Phavichitr, S. Wang, S. Chomto, R. Tantibhaedhyangkul, A. Kakourou, S. Intarakhao, et al., Impact of synbiotics on gut microbiota during early life: a randomized, double-blind study, *Sci. Rep.* 11 (1) (2021) 1–12.
- [39] K. Kaźmierczak-Siedlecka, K. Skonieczna-Żydecka, J. Biliński, G. Roviello, L. F. Iannone, A. Atzeni, et al., Gut microbiome modulation and Faecal microbiota transplantation following allogeneic hematopoietic stem cell transplantation, *Cancers.* 13 (18) (2021) 4665.